

ACR16164

## 2,4,5-Tribromoimidazole

### SECTION 1. IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

|                               |                                                                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 产品说明:<br>Product Description: | 2,4,5-三溴咪唑<br>2,4,5-Tribromoimidazole                                                                                                                                                                                                                              |
| Cat No. :                     | 161640000; 161640050; 161640250                                                                                                                                                                                                                                    |
| CAS No                        | 2034-22-2                                                                                                                                                                                                                                                          |
| Molecular Formula             | C3 H Br3 N2                                                                                                                                                                                                                                                        |
| Supplier                      | <b>UK entity/business name</b><br>Fisher Scientific UK<br>Bishop Meadow Road,<br>Loughborough, Leicestershire LE11 5RG, United Kingdom<br><br><b>EU entity/business name</b><br>Thermo Fisher Scientific<br>Janssen Pharmaceuticaaan 3a, 2440 Geel, Belgium        |
| Emergency Telephone Number    | For information <b>US</b> call: 001-800-227-6701 / <b>Europe</b> call: +32 14 57 52 11<br>Emergency Number <b>US</b> :001-201-796-7100 / <b>Europe</b> : +32 14 57 52 99<br><b>CHEMTREC</b> Tel. No. <b>US</b> :001-800-424-9300 / <b>Europe</b> :001-703-527-3887 |
| E-mail address                | begel.sdsdesk@thermofisher.com                                                                                                                                                                                                                                     |
| Recommended Use               | Laboratory chemicals.                                                                                                                                                                                                                                              |
| Uses advised against          | No Information available                                                                                                                                                                                                                                           |

### SECTION 2. HAZARD IDENTIFICATION

|                                                  |                     |                  |
|--------------------------------------------------|---------------------|------------------|
| Physical State<br>Powder Solid                   | Appearance<br>Beige | Odor<br>Odorless |
| <b>Emergency Overview</b><br>Toxic if swallowed. |                     |                  |

#### Classification of the substance or mixture

|                     |            |
|---------------------|------------|
| Acute Oral Toxicity | Category 3 |
|---------------------|------------|

#### Label Elements



Signal Word

Danger

#### Hazard Statements

H301 - Toxic if swallowed

## 2,4,5-Tribromoimidazole

**Precautionary Statements****Prevention**

P264 - Wash face, hands and any exposed skin thoroughly after handling

P270 - Do not eat, drink or smoke when using this product

**Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor

P330 - Rinse mouth

**Storage**

P405 - Store locked up

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Physical and Chemical Hazards**

None identified.

**Health Hazards**

Toxic if swallowed.

**Environmental hazards**

Contains no substances known to be hazardous to the environment or not degradable in waste water treatment plants.

This product does not contain any known or suspected endocrine disruptors.

**SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS**

| Component               | CAS No    | Weight % |
|-------------------------|-----------|----------|
| 2,4,5-Tribromoimidazole | 2034-22-2 | 97       |

**SECTION 4. FIRST AID MEASURES****Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

**Skin Contact**

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.

**Inhalation**

Remove from exposure, lie down. Remove to fresh air.

**Ingestion**

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. If possible drink milk afterwards.

**Most important symptoms and effects**

No information available.

**Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Notes to Physician**

Treat symptomatically.

**SECTION 5. FIRE-FIGHTING MEASURES****Suitable Extinguishing Media**Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Alcohol resistant foam.**Extinguishing media which must not be used for safety reasons**

## 2,4,5-Tribromoimidazole

No information available.

**Specific Hazards Arising from the Chemical**

Thermal decomposition can lead to release of irritating gases and vapors.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6. ACCIDENTAL RELEASE MEASURES****Personal Precautions**

Ensure adequate ventilation.

**Environmental Precautions**

See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up**

Sweep up and shovel into suitable containers for disposal.

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7. HANDLING AND STORAGE****Handling**

Ensure adequate ventilation.

**Storage**

Keep refrigerated.

**Specific Use(s)**

Use in laboratories

**SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION****Control Parameters****Exposure Controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source.

**Personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) Goggles (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

**SAFETY DATA SHEET****2,4,5-Tribromoimidazole**

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                 |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Skin and body protection</b>        | Long sleeved clothing                                                                                                                                                                                                                                                           |
| <b>Respiratory Protection</b>          | Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly |
| <b>Large scale/emergency use</b>       | In case of insufficient ventilation, wear suitable respiratory equipment                                                                                                                                                                                                        |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br>When RPE is used a face piece Fit Test should be conducted                                                         |
| <b>Hygiene Measures</b>                | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                                                          |
| <b>Environmental exposure controls</b> | No information available.                                                                                                                                                                                                                                                       |

**SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES**

|                                                |                                 |                                          |
|------------------------------------------------|---------------------------------|------------------------------------------|
| <b>Appearance</b>                              | Beige                           |                                          |
| <b>Physical State</b>                          | Powder Solid                    |                                          |
| <b>Odor</b>                                    | Odorless                        |                                          |
| <b>Odor Threshold</b>                          | No data available               |                                          |
| <b>pH</b>                                      | No information available        |                                          |
| <b>Melting Point/Range</b>                     | 208 - 212 °C / 406.4 - 413.6 °F |                                          |
| <b>Softening Point</b>                         | No data available               |                                          |
| <b>Boiling Point/Range</b>                     | No information available        |                                          |
| <b>Flash Point</b>                             | No information available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>                        | No data available               |                                          |
| <b>Flammability (solid,gas)</b>                | No information available        |                                          |
| <b>Explosion Limits</b>                        | No data available               |                                          |
| <b>Vapor Pressure</b>                          | No data available               |                                          |
| <b>Vapor Density</b>                           | No data available               | (Air = 1.0)                              |
| <b>Specific Gravity / Density</b>              | No data available               |                                          |
| <b>Bulk Density</b>                            | No data available               |                                          |
| <b>Water Solubility</b>                        | No information available        |                                          |
| <b>Solubility in other solvents</b>            | No information available        |                                          |
| <b>Partition Coefficient (n-octanol/water)</b> |                                 |                                          |
| <b>Component</b>                               | <b>log Pow</b>                  |                                          |
| 2,4,5-Tribromoimidazole                        | 1.96                            |                                          |
| <b>Autoignition Temperature</b>                | No data available               |                                          |
| <b>Decomposition Temperature</b>               | No data available               |                                          |
| <b>Viscosity</b>                               | No data available               |                                          |
| <b>Explosive Properties</b>                    | No information available        |                                          |
| <b>Oxidizing Properties</b>                    | No information available        |                                          |
| <b>Molecular Formula</b>                       | C3 H Br3 N2                     |                                          |
| <b>Molecular Weight</b>                        | 304.76                          |                                          |

**SECTION 10. STABILITY AND REACTIVITY**

## 2,4,5-Tribromoimidazole

|                                 |                                        |
|---------------------------------|----------------------------------------|
| <b>Stability</b>                | Stable under normal conditions.        |
| <b>Hazardous Reactions</b>      | No information available.              |
| <b>Hazardous Polymerization</b> | No information available.              |
| <b>Conditions to Avoid</b>      | None known.                            |
| <b>Materials to avoid</b>       | Strong oxidizing agents. Strong bases. |

**Hazardous Decomposition Products** Nitrogen oxides (NO<sub>x</sub>). Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen halides.

**SECTION 11. TOXICOLOGICAL INFORMATION**

**Product Information** No acute toxicity information is available for this product

**(a) acute toxicity;**

| Component               | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|-------------------------|-------------------------|-------------|-----------------|
| 2,4,5-Tribromoimidazole | LD50 = 34 mg/kg ( Rat ) |             |                 |

**(b) skin corrosion/irritation;** No data available

**(c) serious eye damage/irritation;** No data available

**(d) respiratory or skin sensitization;**

**Respiratory** No data available  
**Skin** No data available

**(e) germ cell mutagenicity;** No data available

**(f) carcinogenicity;** No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;** No data available

**(h) STOT-single exposure;** No data available

**(i) STOT-repeated exposure;** No data available

**Target Organs** No information available.

**(j) aspiration hazard;** No data available

**Symptoms / effects, both acute and delayed** No information available

**SECTION 12. ECOLOGICAL INFORMATION**

**Ecotoxicity effects** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

| Component               | Freshwater Fish       | Water Flea | Freshwater Algae | Microtox |
|-------------------------|-----------------------|------------|------------------|----------|
| 2,4,5-Tribromoimidazole | LC50: 3.8 - 5.8 mg/L, |            |                  |          |

## SAFETY DATA SHEET

## 2,4,5-Tribromoimidazole

|  |                                                                                                                  |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------|--|--|--|
|  | 96h flow-through<br>(Pimephales promelas)<br>LC50: 7.23 - 8.78 mg/L,<br>96h semi-static<br>(Pimephales promelas) |  |  |  |
|--|------------------------------------------------------------------------------------------------------------------|--|--|--|

**Persistence and Degradability** No information available

**Bioaccumulative Potential** No information available

| Component               | log Pow | Bioconcentration factor (BCF) |
|-------------------------|---------|-------------------------------|
| 2,4,5-Tribromoimidazole | 1.96    | No data available             |

**Mobility in soil** No information available

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors  
**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13. DISPOSAL CONSIDERATIONS

**Waste from Residues/Unused Products** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

**Contaminated Packaging** Empty remaining contents. Dispose of in accordance with local regulations. Do not re-use empty containers.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used.

## SECTION 14. TRANSPORT INFORMATION

Road and Rail Transport

**UN-No** UN2811  
**Proper Shipping Name** Toxic solid, organic, n.o.s.  
**Technical Shipping Name** 2,4,5-Tribromoimidazole  
**Hazard Class** 6.1  
**Packing Group** II

IMDG/IMO

**UN-No** UN2811  
**Proper Shipping Name** Toxic solid, organic, n.o.s.  
**Technical Shipping Name** 2,4,5-Tribromoimidazole  
**Hazard Class** 6.1  
**Packing Group** II

IATA

**UN-No** UN2811  
**Proper Shipping Name** Toxic solid, organic, n.o.s.  
**Technical Shipping Name** 2,4,5-Tribromoimidazole  
**Hazard Class** 6.1  
**Packing Group** II

## 2,4,5-Tribromoimidazole

Special Precautions for User No special precautions required

## SECTION 15. REGULATORY INFORMATION

## International Inventories

X = listed, China (IECSC), Europe (EINECS/ELINCS/NLP), U.S.A. (TSCA), Canada (DSL/NDSL), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), Korea (KECL).

| Component               | The Inventory of Hazardous Chemicals (2015 Edition) | List of dangerous goods GB 12268 - 2012 | TCSI | IECSC | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | AICS | KECL |
|-------------------------|-----------------------------------------------------|-----------------------------------------|------|-------|-----------|------|-----|-------|------|------|------|------|
| 2,4,5-Tribromoimidazole | -                                                   | -                                       | X    | -     | 217-997-5 | -    | -   | -     | -    |      | -    | -    |

## National Regulations

## SECTION 16. OTHER INFORMATION

Revision Date 06-Apr-2024  
Revision Summary Not applicable.

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Disclaimer**

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**End of Safety Data Sheet**